

Stefan Berggren, senior product marketing manager - product strategy short range radio, u-blox, says, "Signals can now easily penetrate walls and obstacles in homes, thereby offering a more robust solution. Bluetooth 5 and BLE Mesh technologies are extensively used for making a home smart. Bluetooth Mesh extends coverage and robustness of a home automation system, as data is propagated via node-to-node hops in a mesh network."

"Luminaires, light switches, sensors, thermostats, blinds and air-conditioners can all be connected in a mesh network. A smartphone can also be part of the Bluetooth Mesh network, and an app can easily control lighting and indoor climate systems."

"Solutions for other mesh tech-

nologies such as Thread, Zigbee and Wirepas Mesh are also applicable for home automation. Mesh networks in a smart home can be connected to the cloud via a capillary gateway, either using Wi-Fi/Ethernet through the home router or via cellular access."

Controlling devices wirelessly

Theodore Wang, marketing director - Latin America and India, EZVIZ (smart home security provider, and consumer and residentially focused subsidiary of Hikvision), says, "All smart home products can be connected to apps and cloud platforms. We utilise wireless technologies extensively in IoT-based home automation. Our smart home products, including smart locks, smart doorbells, smart plugs, gas detectors and wireless pet immune PIR detectors support IFTTT, Apple HomeKit, Google Home and Amazon Alexa. These can be connected to EZVIZ app and cloud platform."

"EZVIZ C6C Wi-Fi pan-tilt camera features smart tracking, which upon detection of movement automatically captures and tracks motion while delivering a notification, an image or a video to your smartphone via EZVIZ app in real time. A built-in microphone and speaker enable two-way audio, allowing you to communicate with people on the other side via the app. It also comes with a privacy mode

design, allowing you to turn the lens off and enjoy some alone time."

The company also creates wireless cameras that can deliver features like human detection, human tracking, face detection, face recognition, passenger flow monitoring and smart sound location.

A cellular module is generally present in 90 per cent of modern alarm panels. Depending on system architecture and installation choice, cellular technology can be used as main communication access to alarm receiving centres and to the Internet, or as backup communication when paired with wired or Wi-Fi IP connectivity to the Internet.

Cellular communication is also used in other remote monitoring and alarm systems used in building automation, such as for access control, alert systems for elevators, smoke detectors and domestic healthcare monitoring applications, among others.

Diego Grassi, senior manager - product strategy, for cellular IoT market development, u-blox, says, "Cellular and low-power wide-area (LPWA) chipsets and modules are designed with a unique feature set in terms of security, anti-tampering and energy-efficiency. These address specific needs of IoT use cases including home automation applications."

"LTE Cat 1, Cat M1 and NB1 technologies with modules such as SARA-R4, SARA-N2 and LARA-R3 are now being used or designed in the most innovative home automation systems. Typical applications are residential and commercial security, building surveillance, alarm systems, home automation smart sensors, smoke detectors, actuators, access control and HVAC systems, domestic health monitoring and assisted



Fig. 2: EZVIZ wireless home automation features

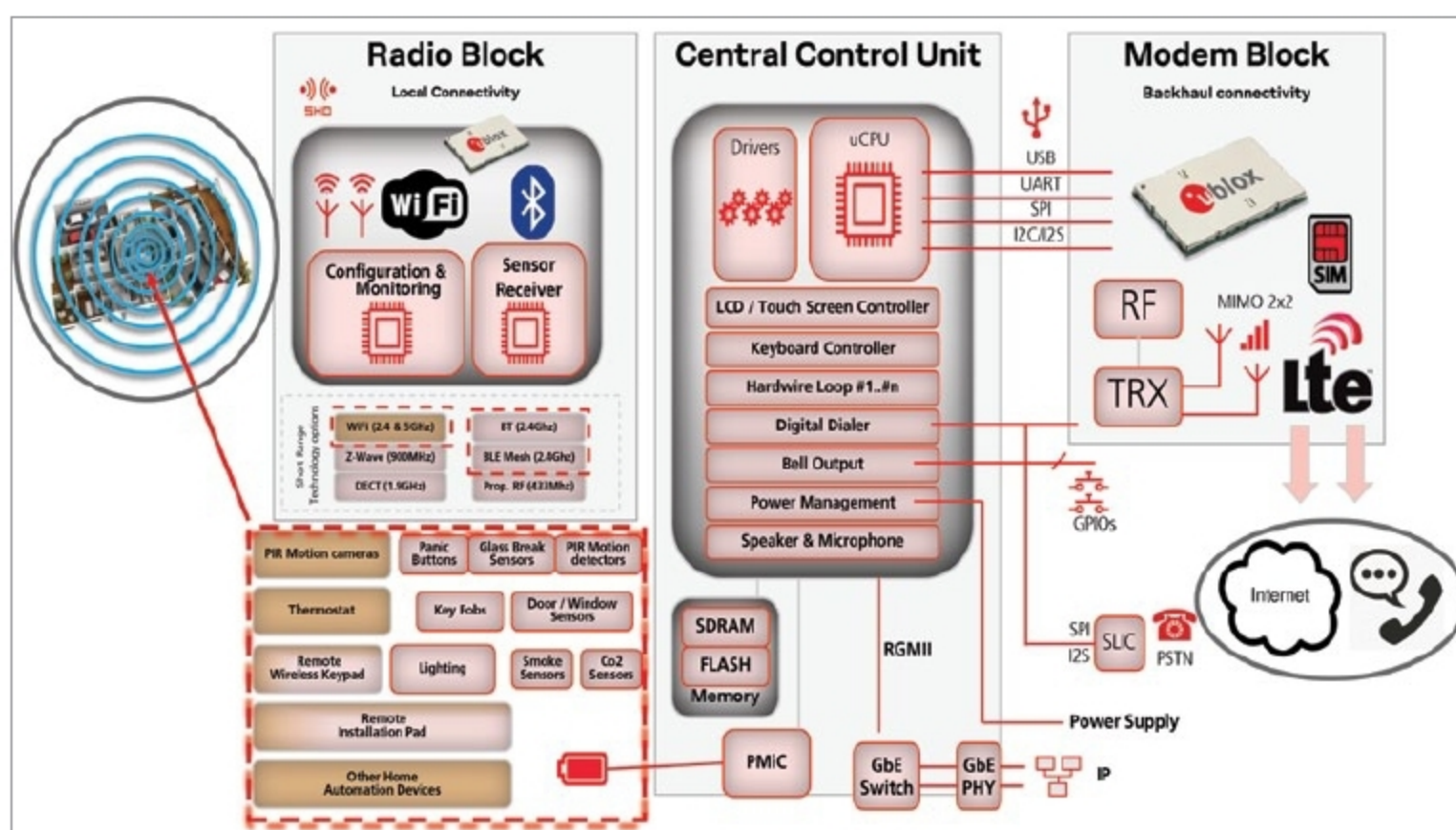


Fig. 3: Wireless connectivity in home automation with u-blox (Credit: u-blox AG)